CAPSTONE PROJECT
PHASE 1 : REVIEW - 2

VAULTSPHERE

A Secure Multi-Tenant SaaS Platform with CI/CD and Auto-Scaling

- **GROUP - 34**
  - Vats pratap singh (22BSA10234)
  - Disha Jadhav (22BSA10061)
  - Navika Mehrotra (22BSA10199)
  - Sourabh Kumrawat(22BSA10112)
  - Keshav Shree (22BSA10129)

SUPERVISOR → **Dr. Virendra singh Khushwah**

REVIEWER → **Dr. Hariharan R.**

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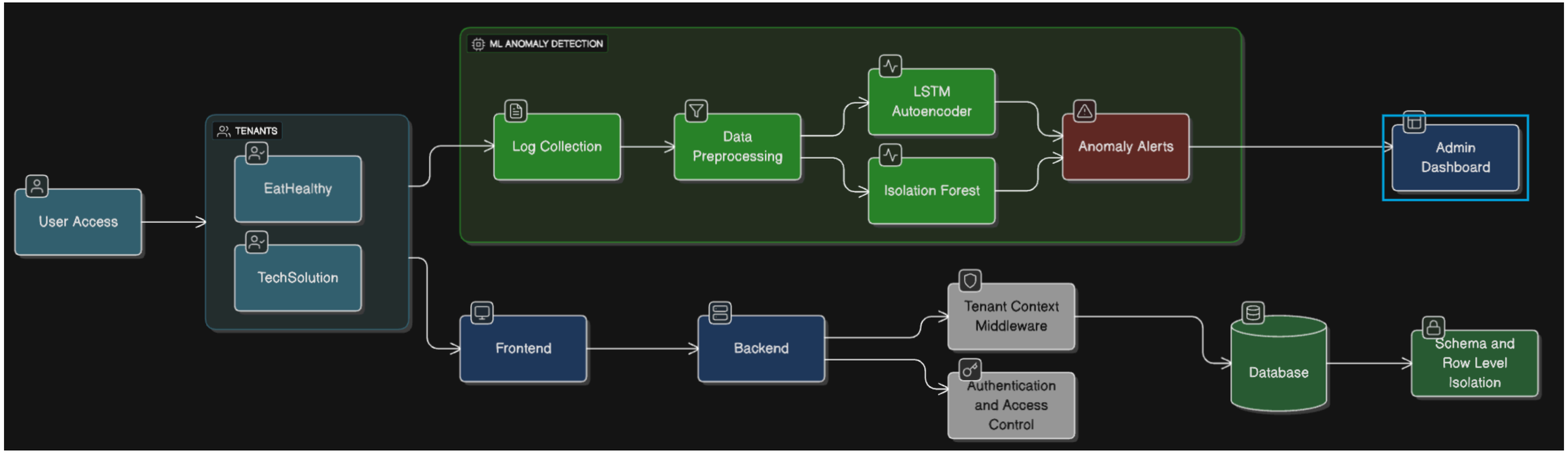
OBJECTIVE

Our primary objective is to design and deploy a robust, cloud-native SaaS platform capable of serving multiple organizations (tenants) from a single codebase, while ensuring complete data security and isolation for each tenant. We aim to streamline software delivery by implementing automated CI/CD pipelines, enabling rapid testing, seamless deployment, and continuous integration of new features. Additionally, the platform will monitor application health in real time, auto-scale resources dynamically to meet varying user demands, and leverage AWS services with Infrastructure as Code (Terraform) to provision and manage scalable, reliable infrastructure efficiently.

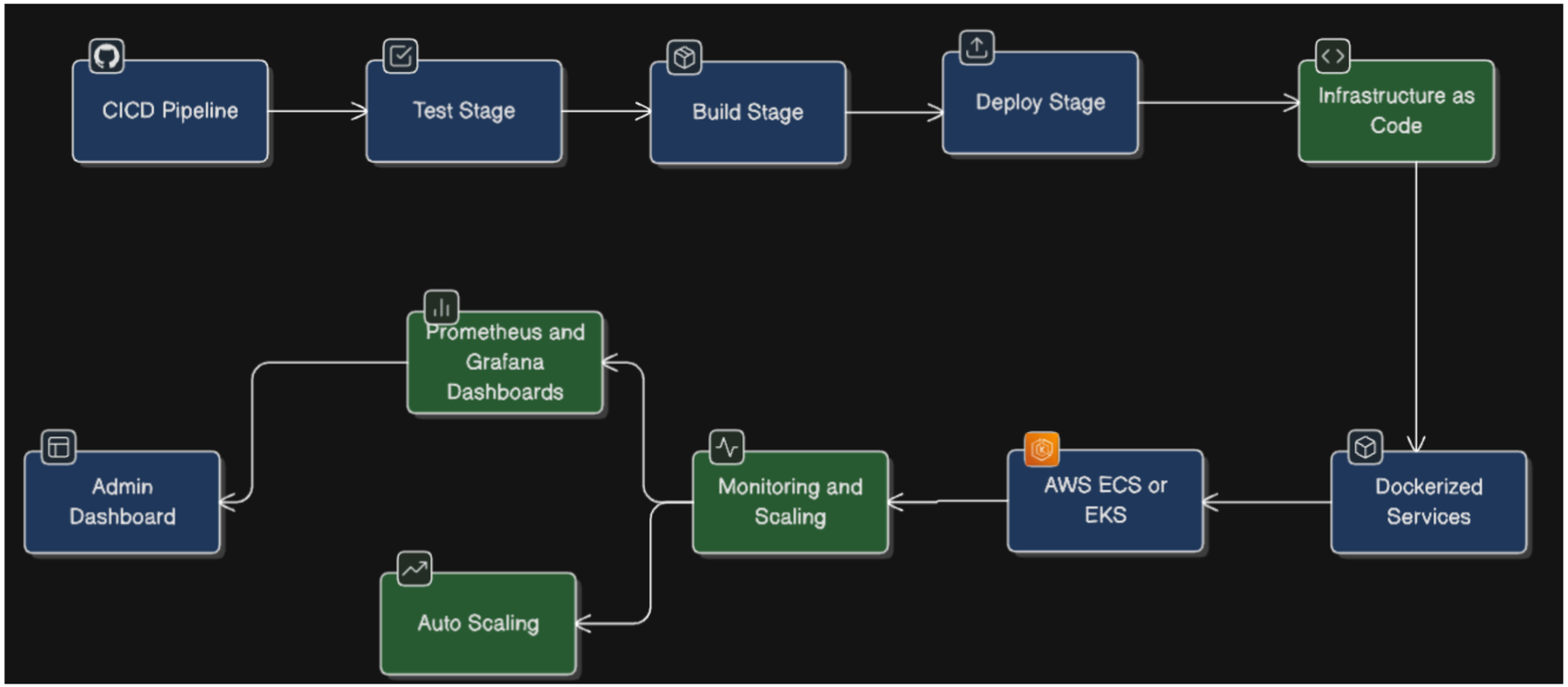
PROBLEM STATEMENT

Companies deploying SaaS solutions often struggle with multiple isolated deployments, which inflate maintenance costs and increase operational complexity. Manual deployment processes introduce delays and errors, making it difficult to scale applications effectively. Poorly designed systems compromise data isolation and access control, creating security vulnerabilities and exposing sensitive client information. Applications fail to adjust dynamically under high usage, resulting in downtime and performance degradation, while monitoring and debugging remain cumbersome without centralized visibility. **VaultSphere addresses these challenges by providing a unified, multi-tenant architecture, automated workflows, and real-time monitoring, ensuring secure, scalable, and efficient operations for all tenants.**

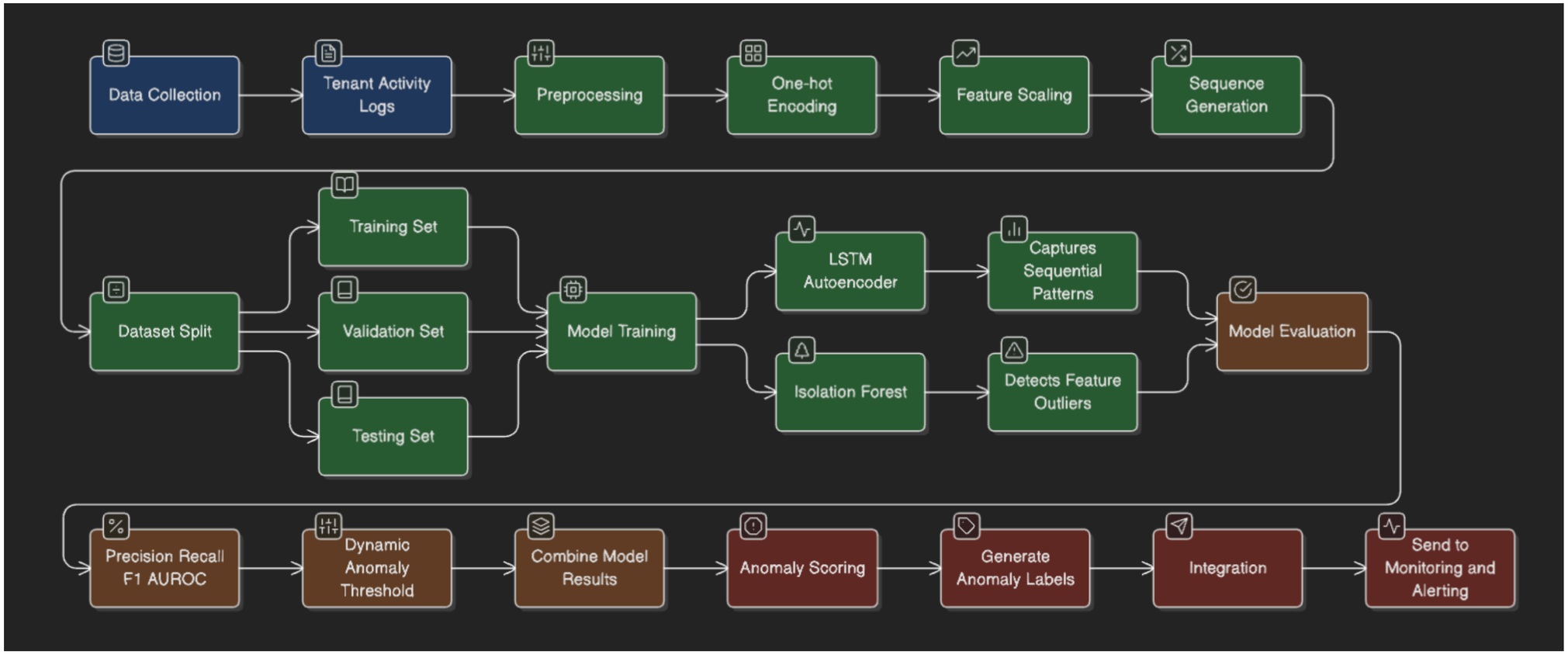
WORKFLOW
Application



WORKFLOW
CI/CD



WORKFLOW
ML


Data Collection
Tenant Activity Logs
Preprocessing
One-hot Encoding
Feature Scaling
Sequence Generation
Dataset Split
Training Set
Validation Set
Testing Set
Model Training
LSTM Autoencoder
Captures Sequential Patterns
Isolation Forest
Detects Feature Outliers
Model Evaluation
Precision Recall F1 AUROC
Dynamic Anomaly Threshold
Combine Model Results
Anomaly Scoring
Generate Anomaly Labels
Integration
Send to Monitoring and Alerting

Proposed Work & Methodology

1. **Frontend Development:** Built modular, responsive, tenant-aware UI using React.js / Next.js.
2. **Backend Development**: Developed RESTful APIs with Express.js and middleware for tenant-level access control.
3. **Database Layer**: Implemented PostgreSQL with schema-based multi-tenancy for secure, organized data storage.
4. **Authentication**: Integrated JWT login mechanisms with potential extension to OAuth 2.0 / AWS Cognito.
5. **CI/CD Pipeline**: Configured GitHub Actions for automated testing, building, and deployment.
6. **Cloud Infrastructure (IaC)**: Provisioned AWS resources using Terraform for scalable, reproducible environments.
7. **Monitoring & Visualization**: Set up Prometheus for metrics collection and Grafana dashboards for real-time insights.
8. **Auto-Scaling**: Implemented dynamic scaling on ECS to handle traffic spikes efficiently.
9. **ML / Anomaly Detection**: Processed tenant activity datasets; trained LSTM Autoencoder for sequential pattern learning and Isolation Forest for outlier detection
10. **Overall Approach**: Ensured secure, scalable, and maintainable multi-tenant SaaS platform from development to production.

Novelty

- **Truly Multi-Tenant Architecture**: Implements a unified backend with tenant-aware routing, maximizing resource utilization while maintaining strict data isolation.
- **End-to-End CI/CD Automation**: Automated pipelines ensure zero-downtime deployments, rapid feature rollouts, and consistent environment management.
- **Cloud-Native Infrastructure**: Uses AWS services (ECS/EKS, S3, RDS) with Terraform IaC, enabling scalable, reproducible, and secure deployments.
- **Built-In Monitoring & Auto-Scaling**: Prometheus + Grafana dashboards and auto-scaling logic allow the platform to self-heal and scale dynamically based on load.
- **Modular Feature Rollouts:** Supports per-tenant customization, enabling flexible and targeted feature delivery without affecting other tenants.
- **ML-Driven Anomaly Detection**: Integrates LSTM & Isolation Forest-based models to proactively detect unusual tenant behavior, enhancing security and reliability.

Real-Time Use Cases

1. **Food Company – EatHealthy**:
   - Each branch registers as a tenant on VaultSphere.
   - Employees log tasks such as inventory updates, order processing, and delivery tracking.
   - LSTM + Isolation Forest-based anomaly detection monitors unusual order patterns or suspicious system activity.
   - Auto-scaling ensures smooth operation during peak hours.
2. **IT Company – TechSolution**:
   - Each department acts as a tenant managing project tasks, code commits, and resource access.
   - VaultSphere provides tenant isolation, ensuring departments see only their own data.
   - Anomaly detection identifies irregular logins, unauthorized resource access, or abnormal API usage.
   - Real-time dashboards visualize performance and anomalies for IT admins.

Tools & Technologies Used

Frontend:

- React.js – Build dynamic, modular, and responsive user interfaces.
- Fetch API – For communicating with backend REST APIs.

Backend:

- Node.js – REST API development, multi-tenant logic, and business rules enforcement.
- JWT – Authentication and secure password management.

Database:

- PostgreSQL – Multi-tenant data storage using schema-level or row-level isolation.

CI/CD & Automation:

- GitHub Actions – Automate testing, builds, and deployments.
- Docker – Containerization for consistent deployments.
- Terraform – Infrastructure as Code (IaC) for AWS resources provisioning

Cloud & DevOps:

- AWS ECS / EKS – Deploy containerized backend services.
- S3 – Static frontend hosting.
- VPC, Security Groups – Networking and security setup.

Monitoring & Logging:

- Prometheus / Grafana – Collect metrics, visualize system performance, and setup alerts.

Literature Review (SaaS)

- **Multi-Tenant Architecture**: Research demonstrates that multi-tenant SaaS platforms significantly enhance resource utilization while ensuring strict tenant data isolation using schema-level or row-level separation.
- **CI/CD in SaaS**: Studies strongly emphasize that automated CI/CD pipelines streamline deployments, reduce downtime, and maintain system consistency, enabling rapid and reliable feature rollouts.
- **Cloud Infrastructure & Scaling**: Prior works on cloud-native SaaS platforms highlight leveraging AWS services like ECS, EKS, RDS, and S3 combined with Infrastructure as Code (Terraform) to achieve secure, reproducible, and scalable deployments.
- **Security & Access Control**: Literature recommends JWT, OAuth 2.0, and AWS Cognito for robust authentication, along with role-based access control, to safeguard tenant data and enforce strict access policies.

Literature Review (ML)

- **Anomaly Detection in SaaS Logs**: Research demonstrates that LSTM networks, autoencoders, and Isolation Forests effectively detect abnormal user behaviors and system events, enabling proactive issue identification.
- **Sequence Modeling**: Studies highlight that LSTM models excel at capturing sequential patterns in time-series log data, allowing accurate detection of deviations from normal behavior.
- **Class Imbalance Handling**: Prior works show that techniques like oversampling, threshold tuning, and hybrid modeling substantially improve detection when anomalies are rare, ensuring robust performance in real-world SaaS systems.
- **Evaluation Metrics**: Literature emphasizes the use of precision, recall, F1-score, and AUROC to quantitatively assess anomaly detection effectiveness, providing reliable benchmarks for model performance.

ML Overview

VaultSphere implements a hybrid machine learning approach to detect anomalies in tenant activity logs from EatHealthy (Food Company) and TechSolution (IT Company). Using features like event_type, status, resource_id, timestamp, and user_id, a LSTM Autoencoder is trained to learn normal sequential patterns in the logs, while an Isolation Forest identifies outliers in the feature space. The hybrid approach combines the strengths of sequence modeling and outlier detection to improve anomaly detection despite class imbalance. Anomalies are flagged based on reconstruction errors and isolation scores, and evaluated using metrics such as Precision, Recall, and F1-Score, ensuring accurate detection of rare abnormal events. The results are integrated into the /ml/anomaly-check API, providing real-time alerts and dashboards for tenant administrators.

ML Methodology

1. **Data Collection**: Tenant activity logs collected from EatHealthy (Food Company) and TechSolution (IT Company), including events like login, resource access, and task updates.
2. **Data Preprocessing:**
    a. One-hot encoding for categorical features (event_type, status, resource_id).
    b. Normalization/scaling for numeric features.
    c. Sequence creation for LSTM using a fixed seq_length.
3. **Modeling Approach:**
    a. LSTM Autoencoder: Learns normal sequential patterns in tenant logs and flags deviations as anomalies.
    b. Isolation Forest: Detects outliers in the feature space using unsupervised ensemble method.
    c. Hybrid Detection: Combine LSTM reconstruction error and Isolation Forest anomaly scores for robust detection.
4. **Handling Class Imbalance:**
    a. Threshold tuning based on validation data.
    b. Oversampling rare anomaly sequences during LSTM training.
5. **Evaluation:**
    a. Metrics: Precision, Recall, F1-Score.
    b. Compare detection performance between LSTM-only, Isolation Forest-only, and hybrid approach.
6. **Integration:** Anomalies are served via /ml/anomaly-check API, feeding real-time dashboards and alerts for tenant admins.

ML Implementation & Results

ML Implementation:

Datasets Used:-

a. **VaultSphere_Food (EatHealthy):**
   i. Synthetic tenant logs meticulously capturing user actions, task updates, and resource access for a food company, enabling anomaly detection in operational workflows.

b. **VaultSphere_IT (TechSolution):**
   i. Synthetic tenant logs for an IT company, with a similar multi-tenant structure, allowing comparison of anomaly patterns across industries.

c. **Synthetic_VaultSphere_Logs:**
   i. Additional simulated multi-tenant events, augmenting model training and testing, ensuring better generalization of anomaly detection.

d. **KDDCup 1999 – 10% Subset:**
   i. A benchmark network intrusion dataset, used for validating model performance on standard anomaly detection tasks.

e. **UNSW-NB15:**
   i. A modern network intrusion dataset, providing realistic, challenging anomalies to test the robustness of the models in near real-world scenarios.

ML Implementation & Results

ML Implementation:

Methodology:-

1. **Preprocessing:**
    a. One-hot encoding for categorical features: event_type, status, resource_id.
    b. Scaling numeric features to normalize input ranges.
    c. Sequence creation for LSTM input, enabling the model to learn temporal patterns in tenant logs.
2. **Models:**
    a. LSTM Autoencoder: Learns sequential patterns in multi-tenant logs to detect anomalous behaviors that deviate from normal activity.
    b. Isolation Forest: Identifies outliers in feature space, complementing LSTM predictions to improve detection coverage.
3. **Training & Validation:**
    a. Dataset split: 70% training, 15% validation, 15% testing on Synthetic_VaultSphere_Logs to ensure proper model generalization.
    b. Early stopping based on validation loss to prevent overfitting during LSTM training.
4. **Evaluation Metrics:**
    a. Precision, Recall, F1-score, and AUROC used to quantitatively assess anomaly detection performance across datasets.

ML Implementation & Results

ML Result:

1. Anomaly Detection Performance Across Tenants and Models

Dataset / Tenant	Model	Accuracy	Precision	Recall	F1-score
EatHealthy (Food)	LSTM Autoencoder	0.86	0.74	0.69	0.71
EatHealthy (Food)	Isolation Forest	0.83	0.70	0.67	0.68
TechSolution (IT)	LSTM Autoencoder	0.87	0.75	0.71	0.73
TechSolution (IT)	Isolation Forest	0.84	0.72	0.68	0.70
Synthetic VaultSphere	LSTM Autoencoder	0.85	0.73	0.70	0.71
Synthetic VaultSphere	Isolation Forest	0.82	0.69	0.65	0.67

2. Overall Anomaly Detection Performance on Synthetic VaultSphere Logs

Model	Accuracy	Precision	Recall	F1-Score
LSTM Autoencoder	0.87	0.76	0.72	0.74
Isolation Forest	0.83	0.71	0.67	0.69
Hybrid (LSTM + IF)	0.88	0.77	0.73	0.75

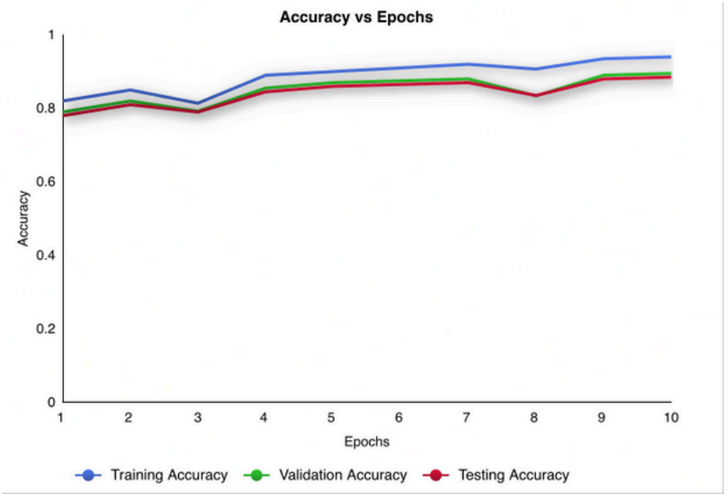
ML Implementation & Results

ML Result:

3. Training, Validation, and Testing Accuracy for Synthetic VaultSphere Logs (LSTM Autoencoder)

Dataset	Model	Training Accuracy	Validation Accuracy	Testing Accuracy
Synthetic VaultSphere	LSTM Autoencoder	0.88	0.84	0.82
Synthetic VaultSphere	Isolation Forest	0.84	0.80	0.78
Synthetic VaultSphere	Hybrid (LSTM + IF)	0.89	0.85	0.83

4. Training, Validation, and Testing Accuracy Curves (Synthetic VaultSphere Logs — LSTM Autoencoder)



ML Implementation & Results

Observations:

- LSTM Autoencoder consistently achieves higher accuracy and AUROC across all VaultSphere datasets, effectively detecting rare anomalies in tenant logs. Threshold tuning is important due to class imbalance.
- Isolation Forest performs slightly lower in recall and F1-score but complements LSTM in low-frequency anomaly detection, making hybrid approaches promising.
- Generalization: Evaluation on synthetic and real tenant datasets shows that models trained on VaultSphere logs generalize well to unseen patterns.
- Performance Curves: Training, validation, and testing losses/metrics demonstrate stable convergence with minimal overfitting, supporting the robustness of the anomaly detection pipeline.

ML Visualization & Insights

Fig. 1 – Food Company: Temporal Distribution of Anomalies

- Peaks correspond to high-order periods and after-hours suspicious activities.

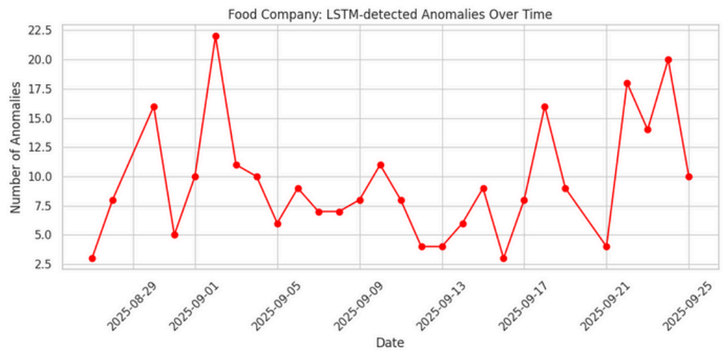


Fig. 2 – IT Solutions: Temporal Distribution of Anomalies

- Higher anomaly frequency during late-night administrative activities and unusual file accesses.

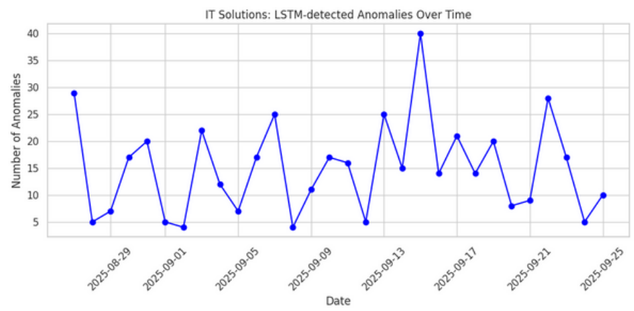


Fig. 3 – Food Company: Event-Type Breakdown of Anomalies

- Most anomalies concentrated in ORDER and PAYMENT events → critical monitoring areas.

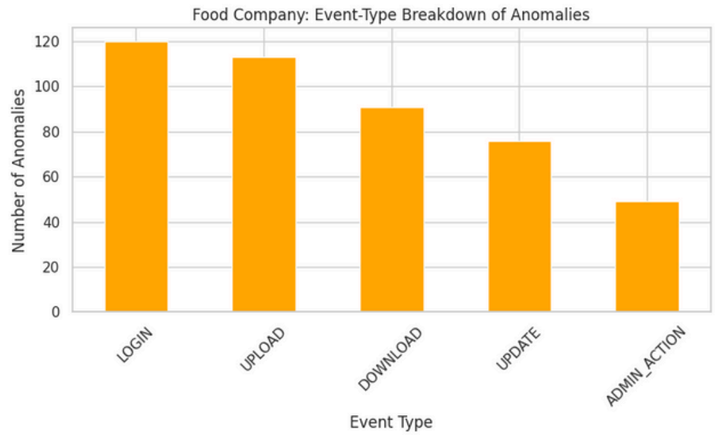
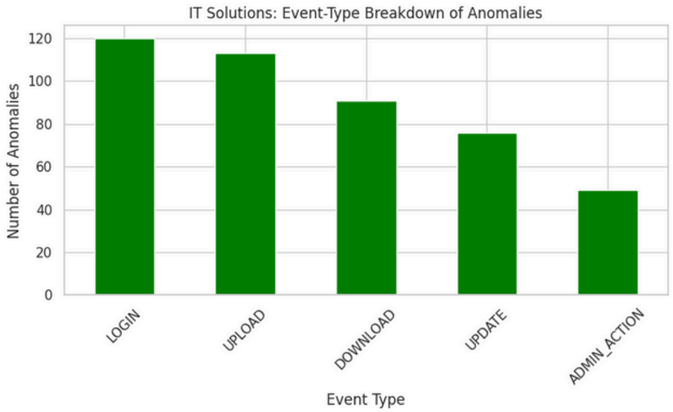


Fig. 4 – IT Solutions: Event-Type Breakdown of Anomalies

- Administrative actions and file uploads dominate anomalies → potential insider threats.



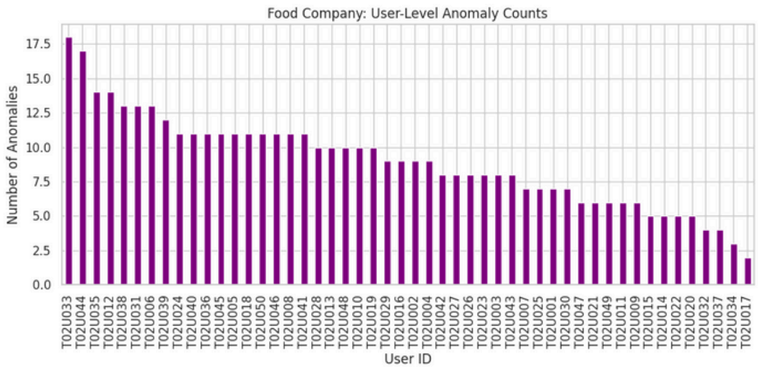
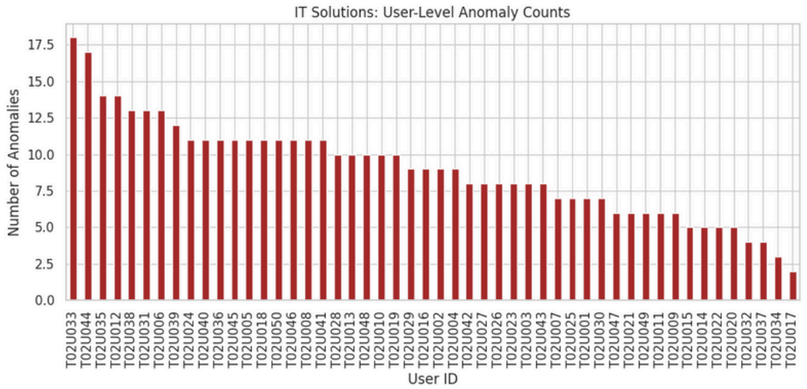


Fig. 5 – Food Company: User-Level Anomaly Counts

- Certain users consistently exhibit higher anomaly counts → unusual work patterns or potential misuse.

Fig. 6 – IT Solutions: User-Level Anomaly Counts

- Some IT users generate a disproportionate number of anomalies → often correlated with after-hours access or bulk operations.



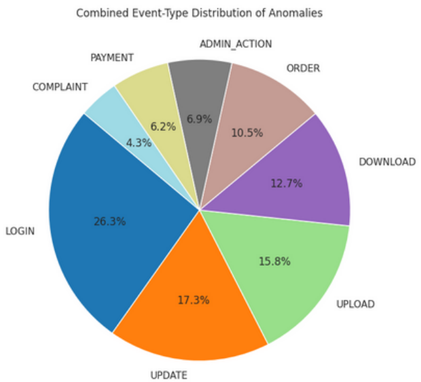


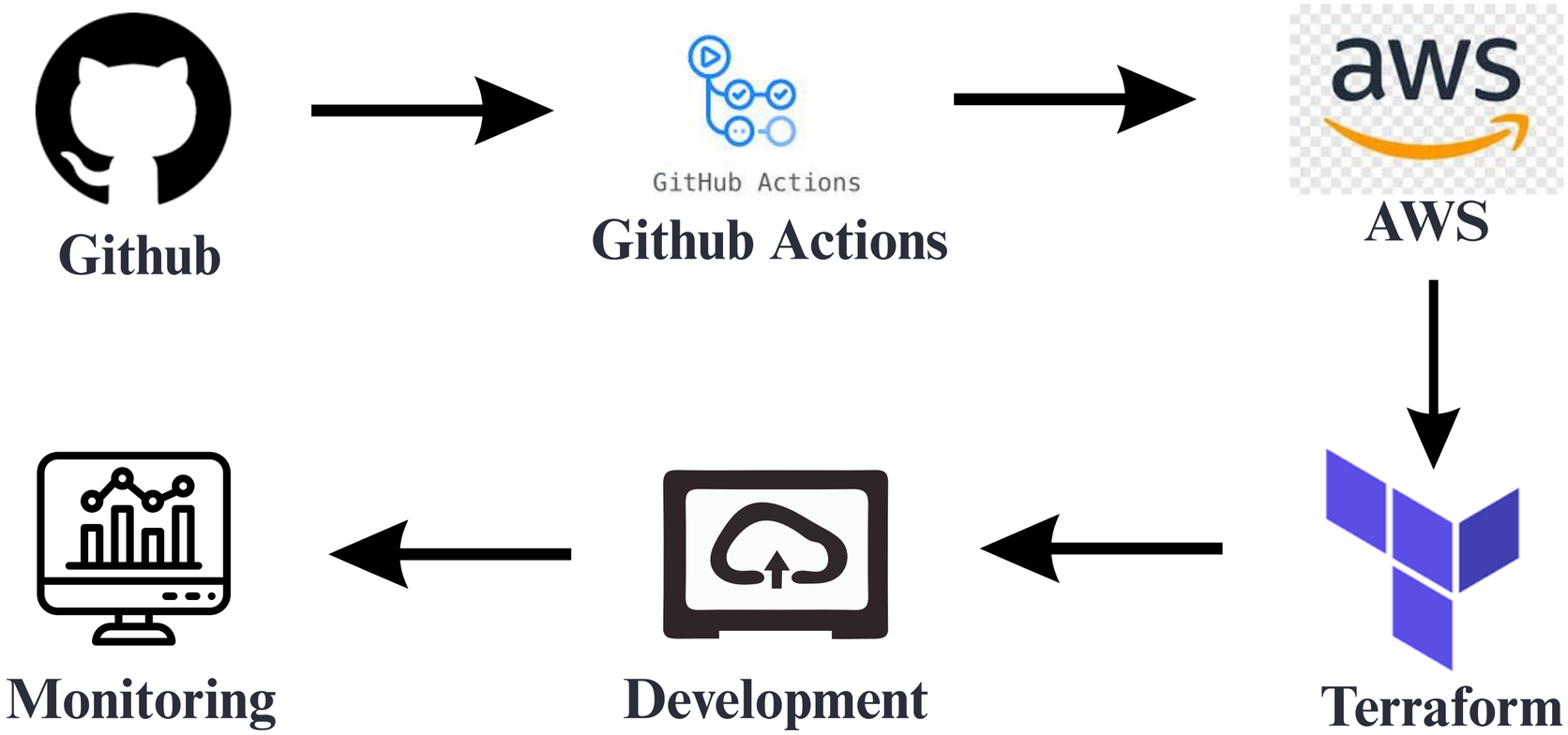
Fig. 7 – Combined Event-Type Distribution Across Both Tenants

- Pie chart showing relative proportions of anomaly types across tenants → highlights major contributing events for targeted security.

CI/CD Pipeline

1. **Version Control**: GitHub repositories manage frontend (React) and backend (Express) code.
2. **Continuous Integration**:
    a. GitHub Actions automatically run linting, unit tests, and build checks on every push.
    b. Ensures code quality and early detection of errors.
3. **Continuous Deployment:**
    a. Successful builds trigger deployment pipelines to AWS ECS/EKS.
    b. Blue-green deployment ensures zero downtime during updates.
4. **Infrastructure as Code (IaC):**
    a. Terraform provisions cloud resources (EC2, RDS, S3, VPC) consistently.
    b. Enables reproducible environments across dev, staging, and production.
5. **Monitoring & Rollback:**
    a. Deployment status monitored via Prometheus + Grafana.
    b. Automatic rollback triggered in case of failed deployment or performance issues.

CI/CD Pipeline



Security Measures

1. **Authentication & Authorization:**
    a. JWT-based authentication for secure access.
    b. Role-based access control (Admin, Tenant User, IT Staff) ensures proper permissions.
2. **Data Isolation:**
    a. Tenant-specific schema/row-level separation in the database.
    b. Prevents cross-tenant data leakage.
3. **Encryption & Secrets Management:**
    a. HTTPS/SSL for secure data transmission.
    b. AWS Secrets Manager stores API keys, database credentials, and other sensitive data.
4. **Monitoring & Alerting:**
    a. Real-time anomaly detection flags suspicious activities.
    b. Prometheus + Grafana dashboards monitor access patterns and alerts.
5. **Rate Limiting & MFA:**
    a. Multi-factor authentication adds extra layer of security.
    b. API rate limiting prevents abuse and brute-force attacks.

Monitoring & Scaling

1. **Performance Monitoring:**
    a. Prometheus collects metrics from backend, frontend, and database.
    b. Grafana dashboards visualize CPU, memory, API latency, and user activity trends.
2. **Log Management:**
    a. Centralized logging of tenant activities and system events.
    b. Supports auditing, troubleshooting, and anomaly detection.
3. **Auto-Scaling:**
    a. ECS/EKS auto-scales based on CPU, memory, and traffic thresholds.
    b. Ensures performance during peak usage without downtime.
4. **Alerts & Notifications:**
    a. Configurable alerts for system anomalies, unusual tenant activity, or resource spikes.
    b. Notifications via email/Slack for real-time response.
5. **Load Balancing:**
    a. Distributes traffic across multiple instances to prevent bottlenecks.
    b. Maintains tenant isolation while scaling horizontally.

Monitoring

```
Vatspratapsingh@MacBook-Air-6 scripts % ./deploy-monitoring.sh
🔍 VaultSphere Stage 8: Enhanced Monitoring & Logging Deployment
==================================================================
[2025-09-26 03:10:37] Starting VaultSphere Enhanced Monitoring Deployment...
[2025-09-26 03:10:37] Checking prerequisites...
[SUCCESS] Prerequisites check passed
[2025-09-26 03:10:38] Creating CloudWatch Log Groups...
[WARNING] Log group /ecs/vaultsphere-backend already exists
[WARNING] Log group /ecs/datadog-agent already exists
[WARNING] Log group /aws/lambda/vaultsphere-request-modifier already exists
[WARNING] Log group /vaultsphere/application already exists
[WARNING] Log group /vaultsphere/security already exists
[WARNING] Log group /vaultsphere/performance already exists
[WARNING] Log group vaultsphere-cloudfront-logs already exists
[WARNING] Log group vaultsphere-cloudfront-api-logs already exists
[2025-09-26 03:10:54] Creating SNS Topics for alerts...
[SUCCESS] Created SNS topic: arn:aws:sns:eu-north-1:687450791196:VaultSphere-Alerts-production
[SUCCESS] Created SNS topic: arn:aws:sns:eu-north-1:687450791196:VaultSphere-Lifecycle-production
[SUCCESS] Created SNS topic: arn:aws:sns:eu-north-1:687450791196:VaultSphere-Security-production
[2025-09-26 03:11:00] Creating IAM roles for enhanced monitoring...

An error occurred (EntityAlreadyExists) when calling the CreateRole operation: Role with name VaultSph
[WARNING] Role may already exist
[SUCCESS] Created IAM roles for enhanced monitoring
[2025-09-26 03:11:02] Creating CloudWatch Dashboards...
{
    "DashboardValidationMessages": []
}
[SUCCESS] Created CloudWatch Dashboard: VaultSphere-production-Overview
[2025-09-26 03:11:04] Creating CloudWatch Alarms...
[SUCCESS] Created CloudWatch Alarms
[2025-09-26 03:11:11] Setting up AWS X-Ray tracing...

usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

  aws help
  aws <command> help
  aws <command> <subcommand> help

aws: error: argument operation: Invalid choice, valid choices are:
```

```
batch-get-traces                          | cancel-trace-retrieval
create-group                              | create-sampling-rule
delete-group                              | delete-resource-policy
delete-sampling-rule                      | get-encryption-config
get-group                                 | get-groups
get-indexing-rules                        | get-insight
get-insight-events                        | get-insight-impact-graph
get-insight-summaries                     | get-retrieved-traces-graph
get-sampling-rules                        | get-sampling-statistic-summaries
get-sampling-targets                      | get-service-graph
get-time-series-service-statistics        | get-trace-graph
get-trace-segment-destination             | get-trace-summaries
list-resource-policies                    | list-retrieved-traces
list-tags-for-resource                    | put-encryption-config
put-resource-policy                       | put-telemetry-records
put-trace-segments                        | start-trace-retrieval
tag-resource                              | untag-resource
update-group                              | update-indexing-rule
update-sampling-rule                      | update-trace-segment-destination
help

[SUCCESS] X-Ray tracing configured
[2025-09-26 03:11:11] Deploying monitoring configuration to backend...
[SUCCESS] Monitoring configuration prepared
[2025-09-26 03:11:12] Testing monitoring setup...
[SUCCESS] Test alert sent successfully
[SUCCESS] Monitoring setup test completed
[SUCCESS] 🎉 Stage 8: Enhanced Monitoring & Logging deployment completed successfully!

📊 Monitoring Resources Created:
  • CloudWatch Log Groups: 8 groups
  • SNS Topics: 3 topics for alerts
  • CloudWatch Dashboard: VaultSphere-production-Overview
  • CloudWatch Alarms: 5 critical alarms
  • IAM Roles: Enhanced monitoring permissions
  • X-Ray Tracing: Service map configured

🔗 Next Steps:
  1. Subscribe email addresses to SNS topics for alerts
  2. Configure Datadog/New Relic integration if needed
  3. Set up custom metrics for business KPIs
  4. Review and adjust alarm thresholds based on baseline
  5. Proceed to Stage 9: Scaling & Performance

📱 Access your dashboard at:
https://console.aws.amazon.com/cloudwatch/home?region=eu-north-1#dashboards:name=VaultS
```

Scaling

```
Vatspratapsingh@MacBook-Air-6 scripts % /Users/Vatspratapsingh/Desktop/VaultSphere/scripts/stage9-deployment-summary.sh
🚀 VaultSphere Stage 9: Scaling & Performance Infrastructure Summary
==========================================================
[2025-09-26 03:20:16] Generating comprehensive deployment summary...

📊 INFRASTRUCTURE OVERVIEW
==================================

🐳 Container Registry (ECR)
Repository: 687450791196.dkr.ecr.eu-north-1.amazonaws.com/vaultsphere-backend
-----------------------------------------------------------------------------------------------------------------
|                                                  DescribeRepositories                                           |
+-----------------------------------+-------------------------------+-------------------------------------------------------------+
|              CreatedAt            |              Name             |                                   URI                                   |
+-----------------------------------+-------------------------------+-------------------------------------------------------------+
|  2025-08-27T23:46:14.649000+05:30 |  vaultsphere-backend          |  687450791196.dkr.ecr.eu-north-1.amazonaws.com/vaultsphere-backend  |
+-----------------------------------+-------------------------------+-------------------------------------------------------------+

🏗️ECS Cluster
-------------------------------------------------------------------------------
|                                DescribeClusters                             |
+----------------+-----------------------+---------------+---------------+----------+
| ActiveServices |         Name          | PendingTasks  | RunningTasks  | Status   |
+----------------+-----------------------+---------------+---------------+----------+
|  1             |  vaultsphere-cluster  |  0            |  0            |  ACTIVE  |
+----------------+-----------------------+---------------+---------------+----------+

🔄 ECS Service
-------------------------------------------------------------------------------
|                                DescribeServices                             |
+---------------+-------------------------------+---------------+---------------+----------+
| DesiredCount  |             Name              | PendingCount  | RunningCount  | Status   |
+---------------+-------------------------------+---------------+---------------+----------+
|  3            |  vaultsphere-backend-service  |  0            |  0            |  ACTIVE  |
+---------------+-------------------------------+---------------+---------------+----------+

⚖️Application Load Balancer
-----------------------------------------------------------------------------------------------------------------
|                                                DescribeLoadBalancers                                            |
+-------------------------------------------------------------+-------------------+-------------------+---------+---------------+
|                          DNSName                            |       Name        |      Scheme       |  State  |     Type      |
+-------------------------------------------------------------+-------------------+-------------------+---------+---------------+
|  vaultsphere-alb-723019547.eu-north-1.elb.amazonaws.com     |  vaultsphere-alb  |  internet-facing  |  active |  application  |
+-------------------------------------------------------------+-------------------+-------------------+---------+---------------+

🎯 Target Group Health

📈 Auto Scaling Configuration
-----------------------------------------------------------------------------------------------------------------
|                                                DescribeScalableTargets                                          |
+---------------+-------------------------------------------------------------------------------------------------+
|  MaxCapacity  |  20                                                                                             |
|  MinCapacity  |  3                                                                                              |
|  ResourceId   |  service/vaultsphere-cluster/vaultsphere-backend-service                                        |
|  RoleARN      |  arn:aws:iam::687450791196:role/aws-service-role/ecs.application-autoscaling.amazonaws.com/AWSServiceRoleForApplicationAutoScaling_ECSService  |
+---------------+-------------------------------------------------------------------------------------------------+

📊 Auto Scaling Policies
-----------------------------------------------------------------------------------------------------
|                                              DescribeScalingPolicies                                |
+-------------------------------------+-------------------------------+-----------------------+--------------+
|             MetricType              |             PolicyName        |       PolicyType      | TargetValue  |
+-------------------------------------+-------------------------------+-----------------------+--------------+
|  ECSServiceAverageCPUUtilization    |  vaultsphere-cpu-scaling-policy     |  TargetTrackingScaling |  70.0        |
|  ECSServiceAverageMemoryUtilization |  vaultsphere-memory-scaling-policy  |  TargetTrackingScaling |  80.0        |
+-------------------------------------+-------------------------------+-----------------------+--------------+

🌐 CloudFront CDN Distribution
-------------------------------------------------------------------------------
|                                ListDistributions                            |
+-------------------------------+----------+----------------+----------------+----------+
|          DomainName           | Enabled  |       Id       |   PriceClass   |  Status  |
+-------------------------------+----------+----------------+----------------+----------+
|  dpyusre67kf57.cloudfront.net |  True    |  ET0D51ZX67FQJ |  PriceClass_All |  Deployed |
+-------------------------------+----------+----------------+----------------+----------+

📡 CloudWatch Alarms

🔗 ACCESS ENDPOINTS
==================================

Application Load Balancer: http://vaultsphere-alb-723019547.eu-north-1.elb.amazonaws.com
CloudFront CDN: https://dpyusre67kf57.cloudfront.net
```

```
📋 DEPLOYMENT DETAILS
==================================

Container Configuration:
• CPU: 1024 units (1 vCPU)
• Memory: 2048 MB (2 GB)
• Health Check: /health endpoint
• Logging: CloudWatch Logs (/ecs/vaultsphere-backend)

Auto Scaling Configuration:
• Min Capacity: 3 instances
• Max Capacity: 20 instances
• CPU Target: 70% utilization
• Memory Target: 80% utilization
• Scale-out Cooldown: 5 minutes
• Scale-in Cooldown: 10 minutes

Load Balancer Configuration:
• Health Check Path: /health
• Health Check Interval: 30 seconds
• Healthy Threshold: 2 consecutive checks
• Unhealthy Threshold: 5 consecutive checks
• Timeout: 5 seconds

CDN Configuration:
• Global Distribution: All edge locations
• HTTPS Redirect: Enabled
• Compression: Enabled for static content
• API Caching: Disabled (/api/* paths)
• Static Caching: 24 hours (/static/* paths)
• Access Logging: Enabled (S3: vaultsphere-cloudfront-logs)

🔧 OPERATIONAL COMMANDS
==================================

Monitor ECS Service:
aws ecs describe-services --cluster vaultsphere-cluster --services vaultsphere-backend-service

View Container Logs:
aws logs tail /ecs/vaultsphere-backend --follow

Check Auto Scaling Activity:
aws application-autoscaling describe-scaling-activities --service-namespace ecs --resource-id service/vaultsphere-cluster/vaultsphere-backend-service

Monitor CloudWatch Metrics:
date: illegal option -- d
usage: date [-jnRu] [-I[date|hours|minutes|seconds|ns]] [-f input_fmt]
            [ -z output_zone ] [-r filename|seconds] [-v[+|-]val[y|m|w|d|H|M|S]]
            [[[[mm]dd]HH]MM[[cc]yy][.SS] | new_date] [+output_fmt]
aws cloudwatch get-metric-statistics --namespace AWS/ECS --metric-name CPUUtilization --dimensions Name=ServiceName,Value=vaultsphere-backend-service Name

Update Service (Rolling Deployment):
aws ecs update-service --cluster vaultsphere-cluster --service vaultsphere-backend-service --force-new-deployment

🎯 PERFORMANCE TESTING
==================================

Load Test ALB Directly:
curl -H 'Host: vaultsphere.com' http://vaultsphere-alb-723019547.eu-north-1.elb.amazonaws.com/health

Load Test via CloudFront:
curl https://dpyusre67kf57.cloudfront.net/health

Stress Test (requires Apache Bench):
ab -n 1000 -c 10 http://vaultsphere-alb-723019547.eu-north-1.elb.amazonaws.com/health

📊 COST OPTIMIZATION
==================================

Current Configuration Cost Estimate (Monthly):
• ECS Fargate (3-20 instances): ~$50-300
• Application Load Balancer: ~$20
• CloudFront (1TB transfer): ~$85
• CloudWatch Logs & Metrics: ~$10
• ECR Storage: ~$5
• Total Estimated: ~$170-420/month

💡 Optimization Tips:
• Use Fargate Spot for non-critical workloads (50% cost reduction)
• Implement intelligent caching to reduce origin requests
• Monitor and adjust auto-scaling thresholds based on actual usage
• Use CloudWatch Insights for log analysis to reduce log retention costs
```

Results

1. Developed a fully multi-tenant SaaS MVP supporting multiple organizations on a single codebase.
2. Implemented secure tenant isolation using row-level/data-level separation.
3. Deployed automated CI/CD pipelines for build, test, and deployment.
4. Enabled cloud-native infrastructure with auto-scaling and monitoring dashboards.
5. Integrated ML-based anomaly detection for tenant activity logs using LSTM + Isolation Forest.

THANK YOU